

DEPARTMENT OF MECHANICAL ENGINEERING

MECHATRONICS & IOT 24MEC51

EXERCISE NO: 2

GURUTHARSAN S	- 24MER011
HARISH KUMAR K K	- 24MER015
MADHANRAJ K	- 24MER028
SADAAGOPAAN T R	- 24MER052

MECHATRONICS & IoT

ASSIGNMENT REPORT – 2

TITLE:

Design and simulate a Smart Refrigerator Monitoring System using DHT11 Sensor. The system monitors internal temperature and humidity conditions to improve food preservation and energy efficiency

Done By:

GURTHARSAN S - 24MER011

HARISH KUMAR K K – 24MER015

MADHANRAJ K – 24MER028

SADAAGOPAAN T R – 24MER052

Project Overview:

The Smart Refrigerator Monitoring System using a DHT11 Sensor is designed to continuously monitor the temperature and humidity inside a refrigerator. Maintaining suitable environmental conditions is important for improving food preservation, reducing food spoilage, and supporting efficient refrigerator operation.

The system uses a DHT11 Temperature and Humidity Sensor to measure the internal conditions. An Arduino Uno acts as the main controller and processes the sensor data. Based on predefined threshold values, the system provides visual and audible alerts.

The system uses:

- Green LED – Normal condition
- Yellow LED – Warning condition
- Red LED + Buzzer – Critical condition

The proposed system can be simulated before implementing it as a physical prototype.

Problem Statement

Food stored inside refrigerators can spoil if the temperature rises above the desired range or if unsuitable humidity conditions occur. Conventional refrigerators may not provide continuous external monitoring or immediate alerts when abnormal conditions develop.

The main problems include:

- Lack of continuous temperature monitoring.
- Difficulty in detecting abnormal temperature conditions quickly.
- Food spoilage due to improper storage conditions.
- Unnecessary energy consumption caused by inefficient operation.
- Lack of immediate warning for critical conditions.
- Manual checking of refrigerator conditions.

Therefore, a smart monitoring system is required to continuously measure temperature and humidity and provide alerts when abnormal conditions occur.

Proposed Solution

The proposed system consists of a DHT11 sensor, Arduino Uno, three LEDs, and a buzzer.

The DHT11 sensor continuously measures temperature and humidity. The sensor readings are sent to the Arduino Uno.

The Arduino compares the measured values with predefined prototype thresholds and determines the refrigerator condition.

Prototype Conditions

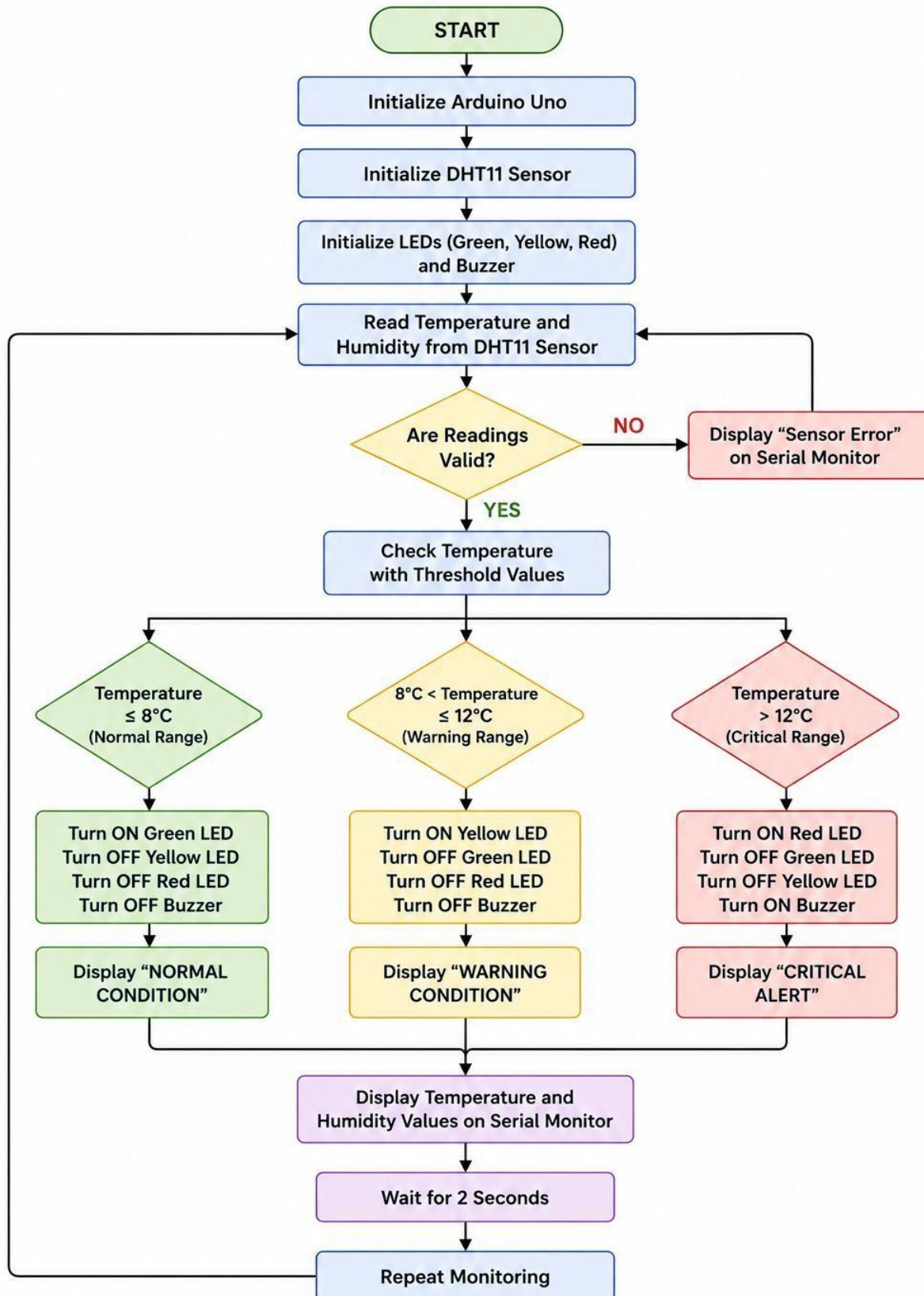
Condition	Temperature	System Output
Normal	2°C–8°C*	Green LED
Warning	8°C–12°C	Yellow LED
Critical	Above 12°C	Red LED + Buzzer

For simulation using a DHT11, use temperature values supported by the simulator/sensor model. Real refrigerator operation near 2–8°C may require a different sensor.

The humidity reading can also be monitored and used as an additional warning parameter.

System Architecture

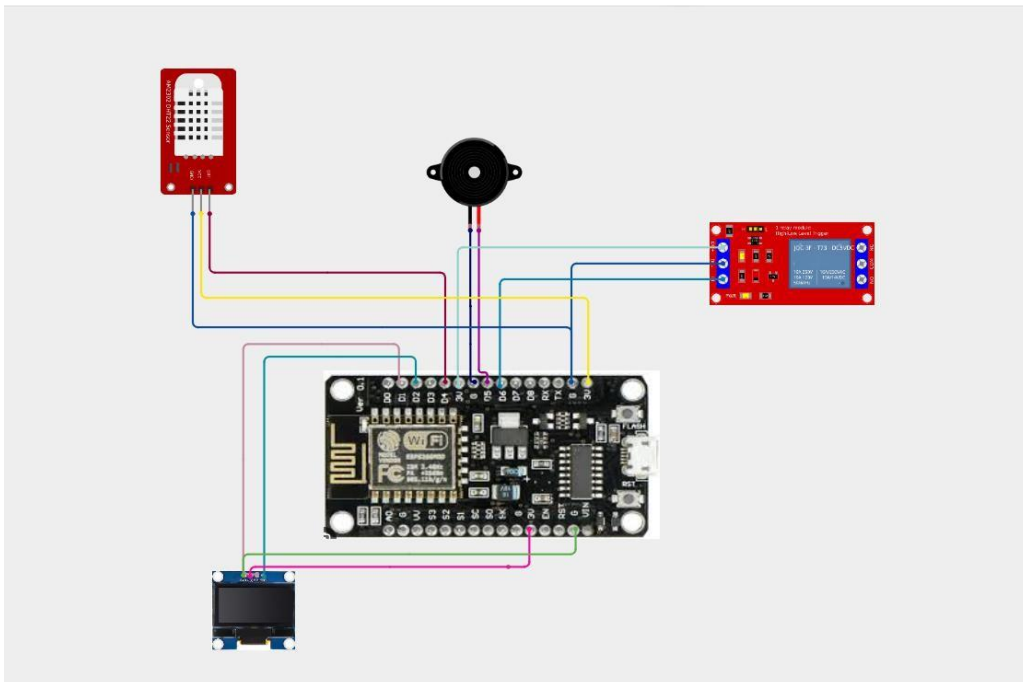
FLOWCHART SMART REFRIGERATOR MONITORING SYSTEM (Using DHT11 Sensor)



Circuit Table

S.No.	Component	Pin/Terminal	Arduino Uno Connection
1	DHT11 Sensor	VCC	5V
2	DHT11 Sensor	GND	GND
3	DHT11 Sensor	DATA	Digital Pin 7
4	Green LED	Anode	Digital Pin 2 through 220 Ω resistor
5	Green LED	Cathode	GND
6	Yellow LED	Anode	Digital Pin 3 through 220 Ω resistor
7	Yellow LED	Cathode	GND
8	Red LED	Anode	Digital Pin 4 through 220 Ω resistor
9	Red LED	Cathode	GND
10	Buzzer	Positive	Digital Pin 5
11	Buzzer	Negative	GND

Circuit Design



Algorithm

1. START
2. Initialize Arduino Uno.
3. Initialize DHT11 Temperature and Humidity Sensor.
4. Initialize Green LED, Yellow LED, Red LED and Buzzer.
5. Read temperature from the DHT11 sensor.
6. Read humidity from the DHT11 sensor.
7. Check whether the sensor readings are valid.
8. Compare the temperature with the predefined threshold values.
9. If the temperature is within the normal range:
 - Turn ON Green LED.
 - Turn OFF Yellow LED.
 - Turn OFF Red LED.
 - Turn OFF Buzzer.
 - Display "NORMAL CONDITION".
10. Else if the temperature is in the warning range:
 - Turn OFF Green LED.
 - Turn ON Yellow LED.
 - Turn OFF Red LED.
 - Turn OFF Buzzer.
 - Display "WARNING CONDITION".
11. Else if the temperature is in the critical range:

Turn OFF Green LED.
Turn OFF Yellow LED.
Turn ON Red LED.
Turn ON Buzzer.
Display "CRITICAL TEMPERATURE ALERT".

12. Display temperature and humidity values.
13. Wait for a short time.
14. Repeat the monitoring process continuously.
15. STOP

Coding

```
/******  
Smart Refrigerator Monitoring System  
ESP8266 NodeMCU + DHT11 + Buzzer + Adafruit IO  
*****/  
  
#include <ESP8266WiFi.h>  
#include "AdafruitIO_WiFi.h"  
#include <DHT.h>  
  
//----- WiFi -----  
#define WIFI_SSID "alexa 007"  
#define WIFI_PASS "7871976757"  
  
//----- Adafruit IO -----  
#define IO_USERNAME "Jr_Frost"  
#define IO_KEY "aio_obVD87ub3BbEG4osjIOQjHNrUn7m"  
  
AdafruitIO_WiFi io(IO_USERNAME, IO_KEY, WIFI_SSID,  
WIFI_PASS);
```



```

// Create Feeds
AdafruitIO_Feed *temperatureFeed = io.feed("temperature");
AdafruitIO_Feed *humidityFeed   = io.feed("humidity");
AdafruitIO_Feed *statusFeed     = io.feed("status");

//----- DHT11 -----
#define DHTPIN 14    // D5 = GPIO14
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

//----- Buzzer -----
#define BUZZER_PIN 12 // D6 = GPIO12

void setup() {

  Serial.begin(115200);
  delay(1000);

  dht.begin();

  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);

  Serial.println("Connecting to Adafruit IO...");

  io.connect();

  while(io.status() < AIO_CONNECTED){
    Serial.print(".");
    delay(500);
  }

  Serial.println();
  Serial.println("Adafruit IO Connected!");
}

void loop() {

```

```
io.run();

float temp = dht.readTemperature();
float hum = dht.readHumidity();

if(isnan(temp) || isnan(hum)){
  Serial.println("DHT Error");
  delay(2000);
  return;
}

String status;

if(temp > 30 || hum > 80){
  status = "ALERT";
  digitalWrite(BUZZER_PIN, HIGH);
}
else if(temp > 20 || hum > 70){
  status = "WARNING";
  digitalWrite(BUZZER_PIN, LOW);
}
else{
  status = "NORMAL";
  digitalWrite(BUZZER_PIN, LOW);
}

Serial.println("-----");
Serial.print("Temperature: ");
Serial.print(temp);
Serial.println(" C");

Serial.print("Humidity: ");
Serial.print(hum);
Serial.println(" %");

Serial.print("Status: ");
Serial.println(status);

// Upload to Adafruit IO
```

```
temperatureFeed->save(temp);  
humidityFeed->save(hum);  
statusFeed->save(status);  
  
delay(5000);  
}
```

System Working

The DHT11 sensor is placed inside the prototype refrigerator environment. It continuously measures the temperature and humidity.

The Arduino Uno reads the sensor values and compares the temperature with the predefined threshold levels.

Normal Condition

When the temperature is within the normal range:

Green LED → ON

This indicates that the refrigerator is operating within the selected prototype range.

Warning Condition

When the temperature increases into the warning range:

Yellow LED → ON

This indicates that the refrigerator condition requires attention.

Critical Condition

When the temperature exceeds the critical threshold:

Red LED → ON

Buzzer → ON

This provides an immediate alert about a potentially unsafe food-storage condition.

The process continues repeatedly to provide continuous monitoring.

Bill of Materials

S.No.	Component	Specification	Quantity	Purpose
1	Arduino Uno	ATmega328P	1	Main Controller
2	DHT11 Sensor	Temperature and Humidity Sensor	1	Environmental Monitoring
3	Green LED	5 mm	1	Normal Indication
4	Yellow LED	5 mm	1	Warning Indication
5	Red LED	5 mm	1	Critical Indication
6	Resistor	220 Ω	3	LED Current Limiting
7	Buzzer	5 V	1	Audible Alert
8	Breadboard	Standard	1	Circuit Assembly
9	Jumper Wires	Male-Male	1 Set	Circuit Connections
10	USB Cable / Power Supply	5 V	1	System Power Supply

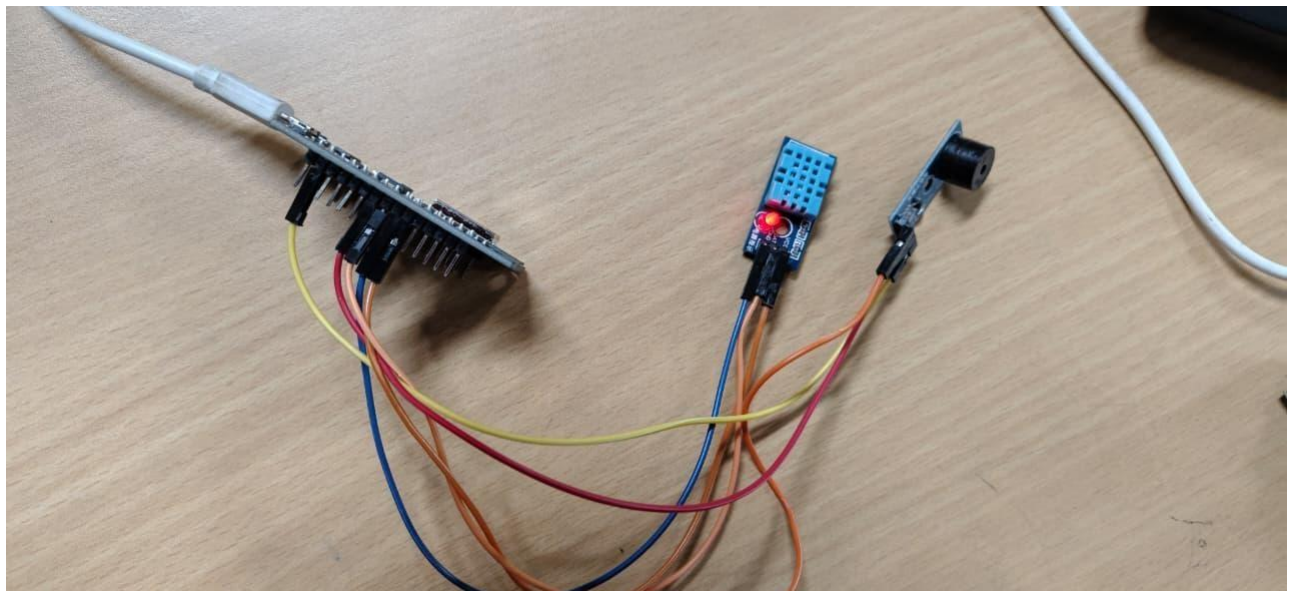
Prototype Implementation

The prototype is assembled using an Arduino Uno, DHT11 sensor, LEDs, buzzer, breadboard, and jumper wires.

The DHT11 sensor is placed inside a model refrigerator or enclosed container. The Arduino and indicator circuit can be placed outside for easy observation.

After uploading the program, the system continuously monitors the temperature and humidity.

Different environmental conditions can be simulated by changing the sensor values in software or by changing the conditions around the prototype.



Results & Performance Analysis

The system can classify the monitored condition into three levels.

Test	Temperature	Humidity	Condition	Output
1	5°C	50%	Normal	Green LED
2	7°C	55%	Normal	Green LED
3	8°C	60%	Normal	Green LED
4	9°C	62%	Warning	Yellow LED
5	11°C	65%	Warning	Yellow LED
6	12°C	68%	Warning	Yellow LED
7	13°C	70%	Critical	Red LED + Buzzer
8	15°C	75%	Critical	Red LED + Buzzer

Performance Analysis

The proposed system continuously monitors environmental conditions and provides immediate visual and audible alerts when the measured temperature crosses predefined thresholds.

The system is:

- Simple to design.
- Low-cost.
- Suitable for simulation.
- Suitable for educational prototypes.
- Expandable with IoT technology.

For a real refrigerator, the DHT11 should be replaced or supplemented with a sensor suitable for low-temperature operation, and actual temperature limits should be selected based on the type of food being stored.

```
-----
Temperature: 28.50 C
Humidity: 73.50 %
Status: WARNING
-----
Temperature: 28.50 C
Humidity: 72.90 %
Status: WARNING
-----
Temperature: 28.60 C
Humidity: 72.60 %
Status: WARNING
-----
Temperature: 28.70 C
Humidity: 76.60 %
Status: WARNING
-----
Temperature: 28.90 C
Humidity: 79.90 %
Status: WARNING
-----
Temperature: 29.20 C
Humidity: 82.60 %
Status: ALERT
-----
Temperature: 29.50 C
Humidity: 84.40 %
Status: ALERT
-----
Temperature: 29.70 C
Humidity: 86.00 %
Status: ALERT
-----
Temperature: 29.90 C
Humidity: 87.40 %
Status: ALERT
-----
Temperature: 30.10 C
Humidity: 88.80 %
Status: ALERT
-----
- - - - -
```

